ACCT102 Management Accounting

Week 10

Master Budget

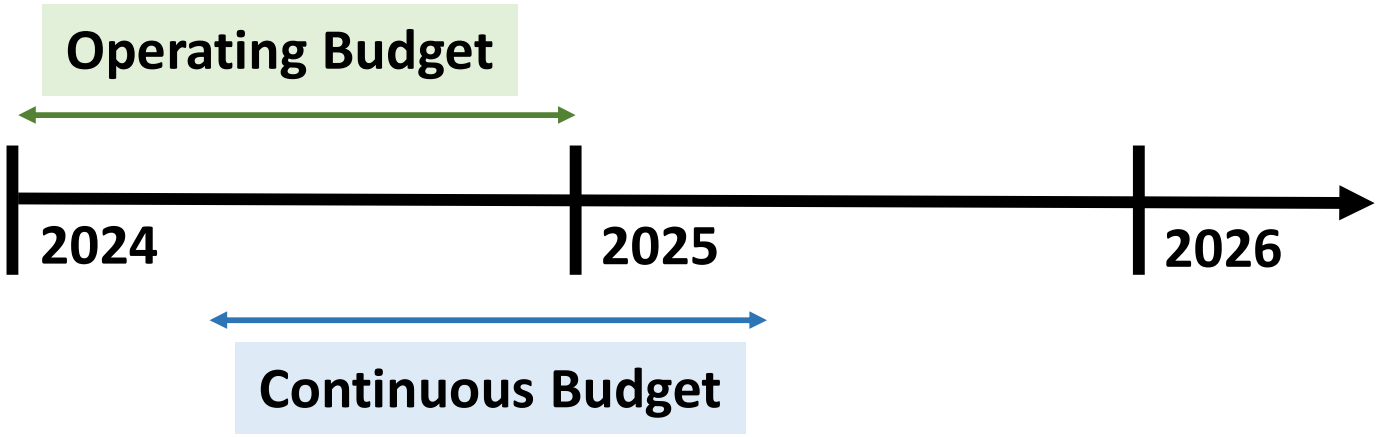
Adapted from © McGraw-Hill Education for ACCT102 (BT)

Learning Objective

Understand why organizations budget and the processes they use to create budgets.

What is a Budget?

- A **budget** is a detailed quantitative plan for acquiring and using financial and other resources over a specified forthcoming time period.



Operating budgets ordinarily cover a one-year period corresponding to a company's fiscal year. Many companies divide their annual budget into four quarters.

A **continuous / perpetual** budget is a 12-month budget that rolls forward one month (or quarter) as the current month (or quarter) is completed.

Key Purposes of a Budget

1. PLANNING - Involves developing objectives and preparing various budgets to achieve those objectives.

Define goals and objectives	Think and plan for the future
Communicate goals and plans	Coordinate activities
Means of allocating resources	Uncover potential bottlenecks

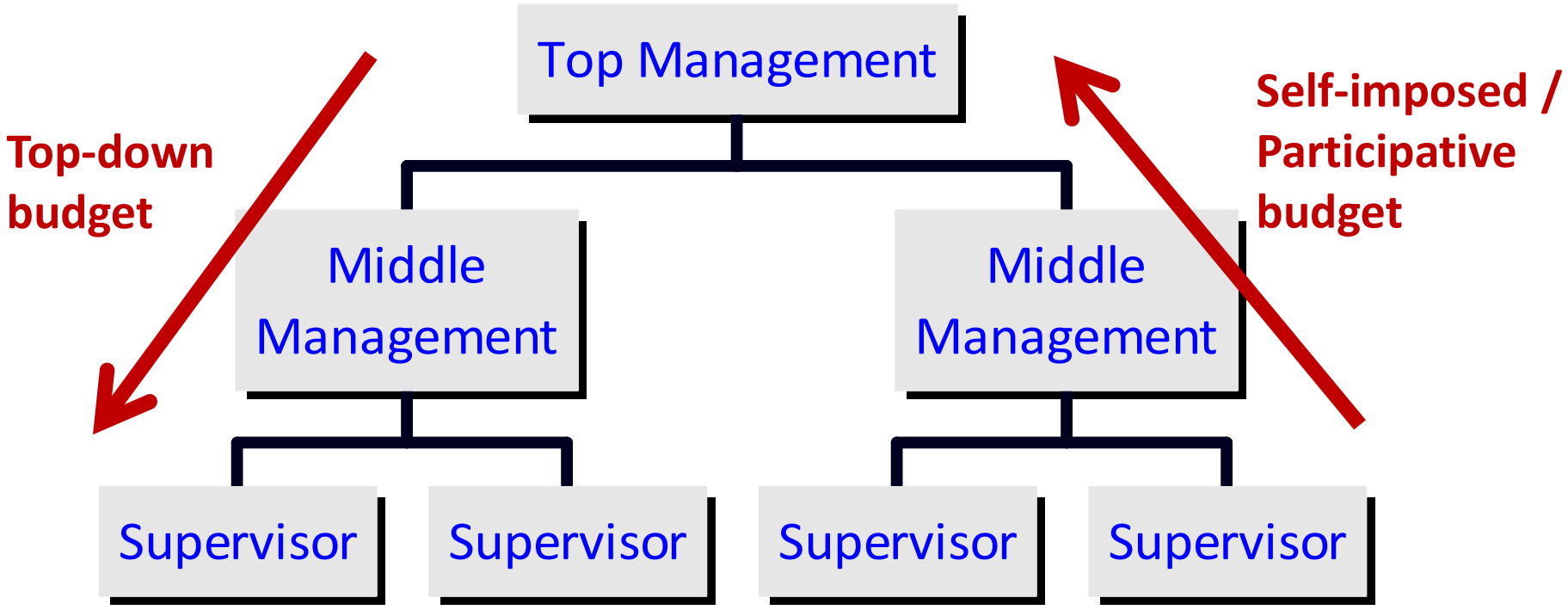
2. CONTROL - Compare budget with actual results. Involves the steps taken by management to increase the likelihood that the objectives set are attained and that all parts of the organization are working together toward that goal.

Improve the efficiency and effectiveness of operations	Evaluate and reward employees

How Do Organizations Create Budgets?

Companies usually create budgets by relying on some combination of top-down budgeting and self-imposed budgeting.

- Broad guidelines issued in terms of overall profits or sales.
- Lower-level managers are directed to prepare budgets that meet those targets.
- Establish challenging targets that are also highly achievable.



Self-imposed Budgets

- It shows **respect** for their opinions when lower-level managers are involved in the budgeting process.
- Budget estimates prepared by front-line managers are often **more accurate** than estimates prepared by top managers.
- **Motivation is generally higher** when individuals participate in setting their own goals than when the goals are imposed from above.
- **It empowers them to take ownership** of the budget and to be accountable for deviations from it.

Other Budgetary Considerations

Budgetary Slack

- Practice of overestimating expenses and/or underestimating revenues to create a “cushion”.
- Occurs when the company employs self-imposed budgets.
- Self-imposed budgets must be reviewed by higher levels of management to minimize this.

Budget Lapsing

- Amounts not spent will lapse in the current period – cannot be brought to the next period.
- Might promote unnecessary spending using unspent amounts towards the end of the period.

Incremental vs Zero-Based Budgets

Incremental Budget

- Begins with a base (usually current year's core budget) **and make incremental changes**.
- Reviews **focus on incremental changes** and may ignore inefficiencies in the core budget

Zero-based budgeting (ZBB)

- Assumes that the company is in its **first year of operation** – no reference to a base year.
- Results in a **fresh consideration of the validity of budget amounts** and the budgeted figures have to be justified from basic
- Motivates managers to **eliminate inefficient expenses**
- More **time consuming**, hence more expensive

Incremental vs Zero-Based Budgets – Example

For example, you own a supermarket store.

Incremental Budget

	Current Year	Budget
Sales	$10m	$11m
# of cashiers	20	21

+10% YoY

Zero-Based Budget

	Budget
Sales	$12m
# of cashiers	20

Based on market / industry study

Based on budget sales, operating hours, expected footfall

The Master Budget

Sales and Marketing Department

- How much sales revenue will we earn?
- How much cash will we collect from customers?
- How much selling expense will we incur and how much cash will we pay related to those expenses?

Operations/Production Department

- How much do we produce?
- How much manufacturing costs will we incur?
- How much cash will we pay to our direct laborers, and manufacturing overhead resources?

Purchasing/Procurement Department

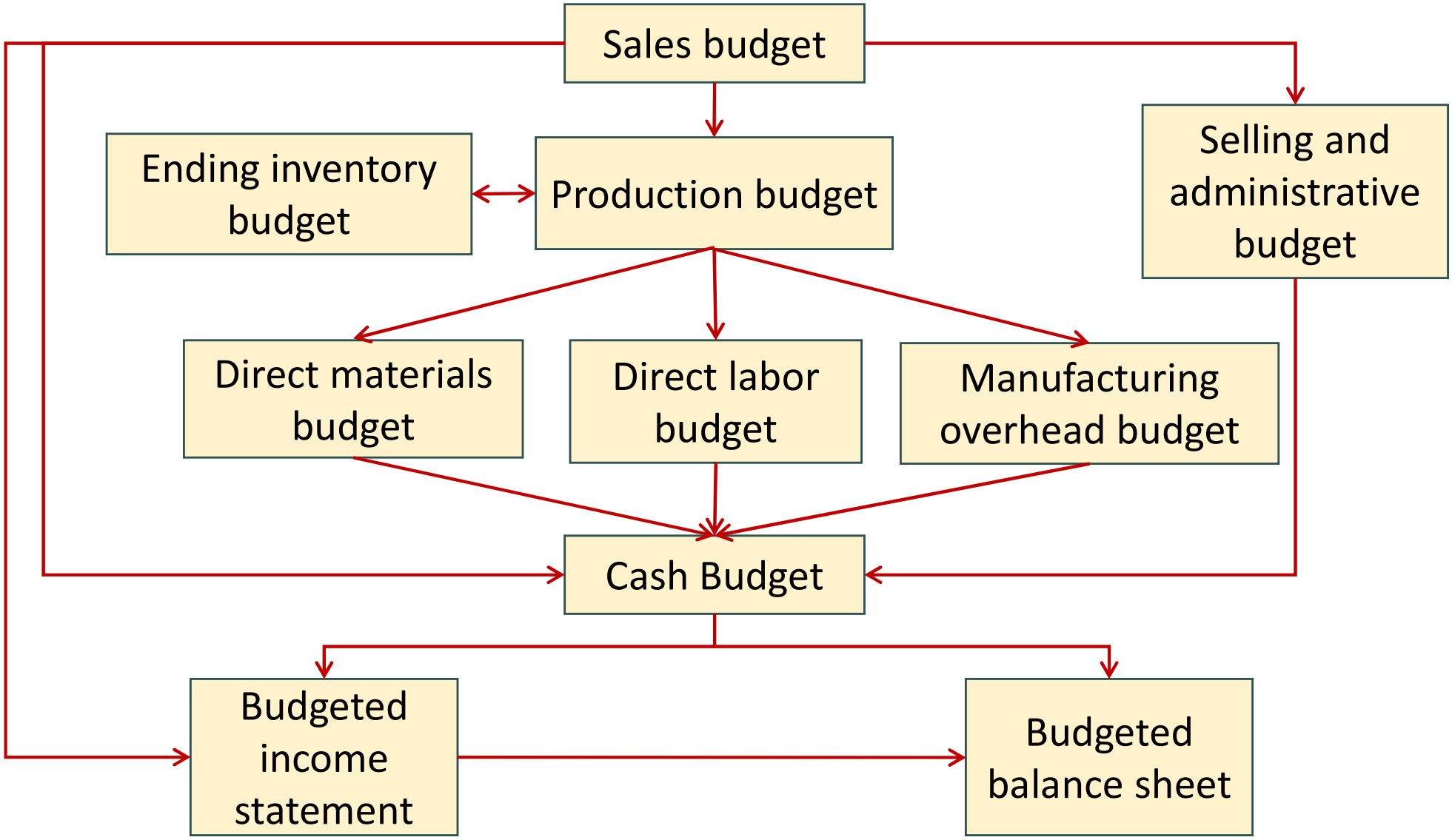
- How much raw material will we need to purchase?
- How much cash will we pay to our suppliers?

Finance Department

- What is the total cost that transferred from FG inventory to COGS?
- How much admin expense will we incur and how much cash will we pay?
- Do we have enough liquidity?
- How much money will we borrow/ repay to lenders – including interest?
- How much operating income will we earn?
- What will our balance sheet look like at the end of the budget period?

Adapted from © McGraw-Hill Education for ACCT102 (BT)

The Master Budget



Adapted from © McGraw-Hill Education for ACCT102 (BT)

Learning Objective

Prepare a sales budget, including a schedule of expected cash collections.

The Sales Budget - Example

Royal Company is preparing a master budget for the **quarter ending June 30th**.

- Budgeted sales for the next five months are:

April	***20,000 units***
May	***50,000 units***
June	***30,000 units***
July	***25,000 units***
August	***15,000 units***

- The selling price is $10 per unit.

The Sales Budget - Example

	April	May	June	Quarter
Budget sales in units				
Selling price per unit				
Total budgeted sales				

Expected Cash Collections - Example

- All sales are on account.
- Royal's collection pattern is:

70% collected in the month of the sale
30% collected in the month following the sale

- As at March 31st, there is an accounts receivable balance of $30,000 that will be collected in full in April.

Expected Cash Collections - Example

	April	May	June	Quarter
Past sales to be collected				
April sales				
May sales				
June sales				
Expected cash collection				

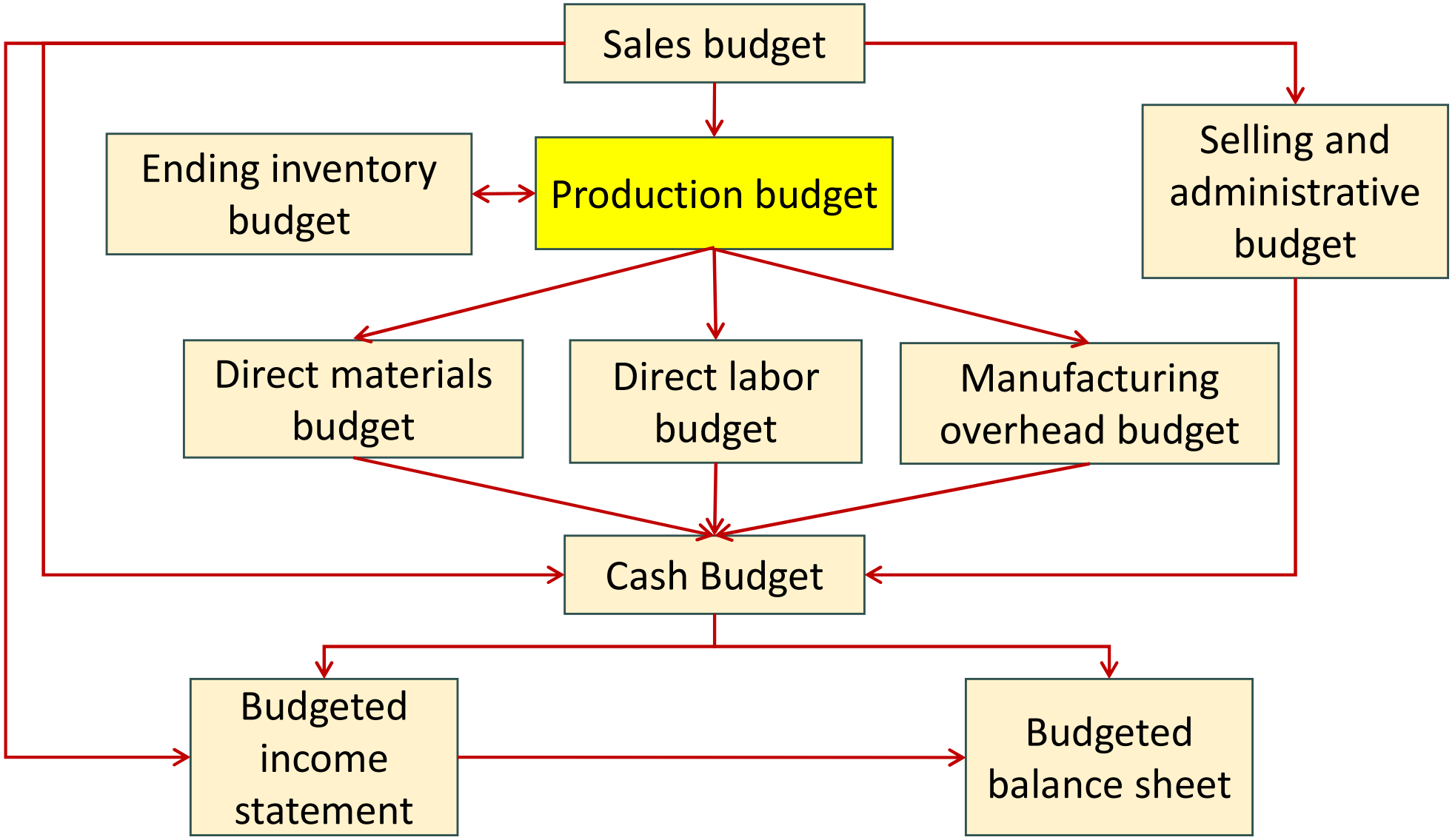
Accounts receivable, at end June =

Quick Check 10.1

What will be the total cash collections for the quarter?

a. $1,030,000
b. $1,000,000
c. $360,000
d. $940,000

The Master Budget



Adapted from © McGraw-Hill Education for ACCT102 (BT)

Learning Objective

Prepare a production budget.

The Production Budget

Completed

Sales Budget and Expected Cash Collections



Production Budget

The production budget must be adequate to meet budgeted sales and to provide for the desired ending inventory.

- *How many units you need?*
- *How many units you have?*
- *How many units to produce?*

The Production Budget - Example

- The management at Royal Company wants ending inventory to be equal to ***20%*** of the following month's budgeted sales in units.
- On March 31st, 4,000 units were on hand.
- July sales quantity: 25,000 units

The Production Budget - Example

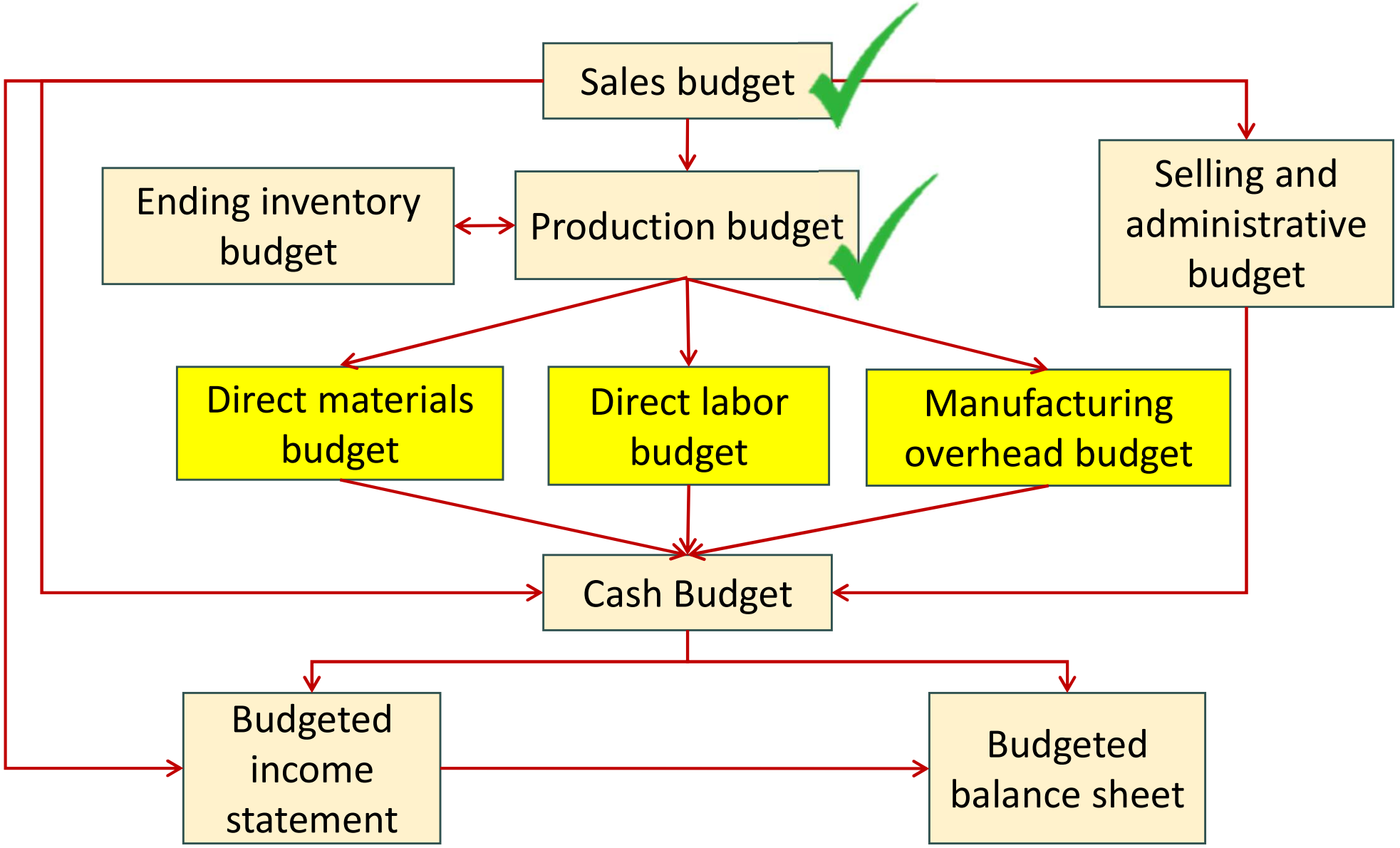
	April	May	June	Quarter
Budget sales in units				
Add: Desired ending inv.				
Total needs				
Less: Begin. Inventory				
Required production				

Quick Check 10.2

What is the required production for the quarter?

a. 100,000 units
b. 105,000 units
c. 101,000 units
d. 96,000 units

The Master Budget



Learning Objective

Prepare a direct materials budget, including a schedule of expected cash disbursements for purchases of materials.

The Direct Materials Budget

Completed

Production Budget → **Direct Materials Budget**

The direct materials budget must be adequate to meet budgeted production and to provide for the desired ending inventory.

- *How much you need?*
- *How much you have?*
- *How much to purchase?*
- *How much cash do you need for the materials?*

Adapted from © McGraw-Hill Education for ACCT102 (BT)

The Direct Materials Budget - Example

- At Royal Company, ***five pounds*** of material are required per unit of product.
- Management wants materials on hand at the end of each month equal to ***10%*** of the following month's production.
- On March 31, 13,000 pounds of material are on hand.

The Direct Materials Budget - Example

	April	May	June	Quarter
Production (units)				
Materials per unit (pounds)				
Production needs				
Add: Desired ending inv.			11,500	
Total needs				
Less: Begin. Inventory				
DM to be purchased				

Quick Check 10.3

How much materials should be purchased for the quarter?

a. 503,500 pounds
b. 505,000 pounds
c. 516,500 pounds
d. 492,000 pounds

Expected Cash Disbursement for DM - Example

- Royal pays $0.40 per pound for its materials.
- Payment schedule is as follows:

  One-half of a month's purchases is paid for in the month of purchase

  The other half is paid in the following month

- The March 31 accounts payable balance is $12,000 to be paid in April.

Expected Cash Disbursement for DM - Example

	April	May	June	Quarter
Past purchases to be paid				
April purchases				
May purchases				
June purchases				
Expected cash disbursement				

Accounts payables, at end June =

Learning Objective

Prepare a direct labor budget.

The Direct Labor Budget

Completed

Production Budget



Direct Labor Budget

- *How much labor is required?*
- *How much does the labor cost?*

The Direct Labor Budget - Example

- At Royal, each unit of product requires 3 minutes of direct labor (0.05 direct labor hours). The labor can be unskilled because the production process is relatively simple and formal training is not required.
- Royal pays its workers at the rate of $10 per hour.
- Royal pays the workers the wages owed at the end of each month.

The Direct Labor Budget - Example

	April	May	June	Quarter
Production (units)				
DLH per unit				
DLH required				
Hourly wage				
Total DL costs				

Learning Objective

Prepare a manufacturing overhead budget.

Manufacturing Overhead Budget

Completed

Production Budget



Manufacturing Overhead Budget

Manufacturing Overhead Budget - Example

- At Royal, manufacturing overhead is applied to units of product on the basis of direct labor hours.
- The variable manufacturing overhead rate is $20 per direct labor hour.
- Fixed manufacturing overhead is $50,000 per month, which includes $20,000 of depreciation of plant assets.
- Costs are paid in the month that it is incurred.

Manufacturing Overhead Budget - Example

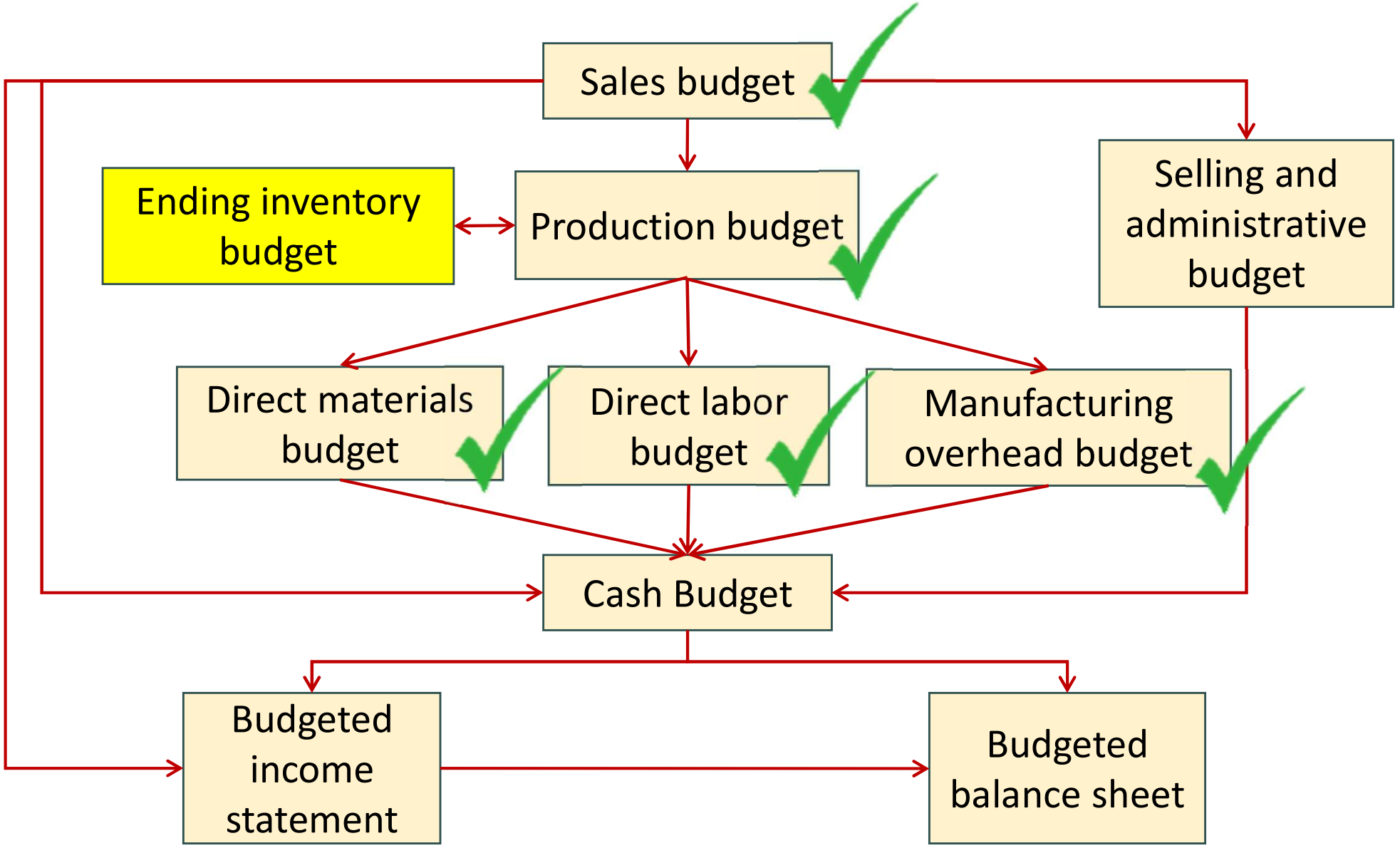
	April	May	June	Quarter
Budget DLH				
Var MOH rate				
Var MOH costs				
Fixed MOH costs				
Total MOH costs				
Less: non-cash costs				
Cash disbursement for MOH				

Quick Check 10.4

What are the total cash disbursements for MOH for the quarter?

a. $251,000
b. $191,000
c. $101,000
d. $150,000

The Master Budget



Learning Objective

Prepare an ending finished goods budget.

Ending Finished Goods Inventory Budget

Completed

Production Budget



Ending Inventory (FG) Budget

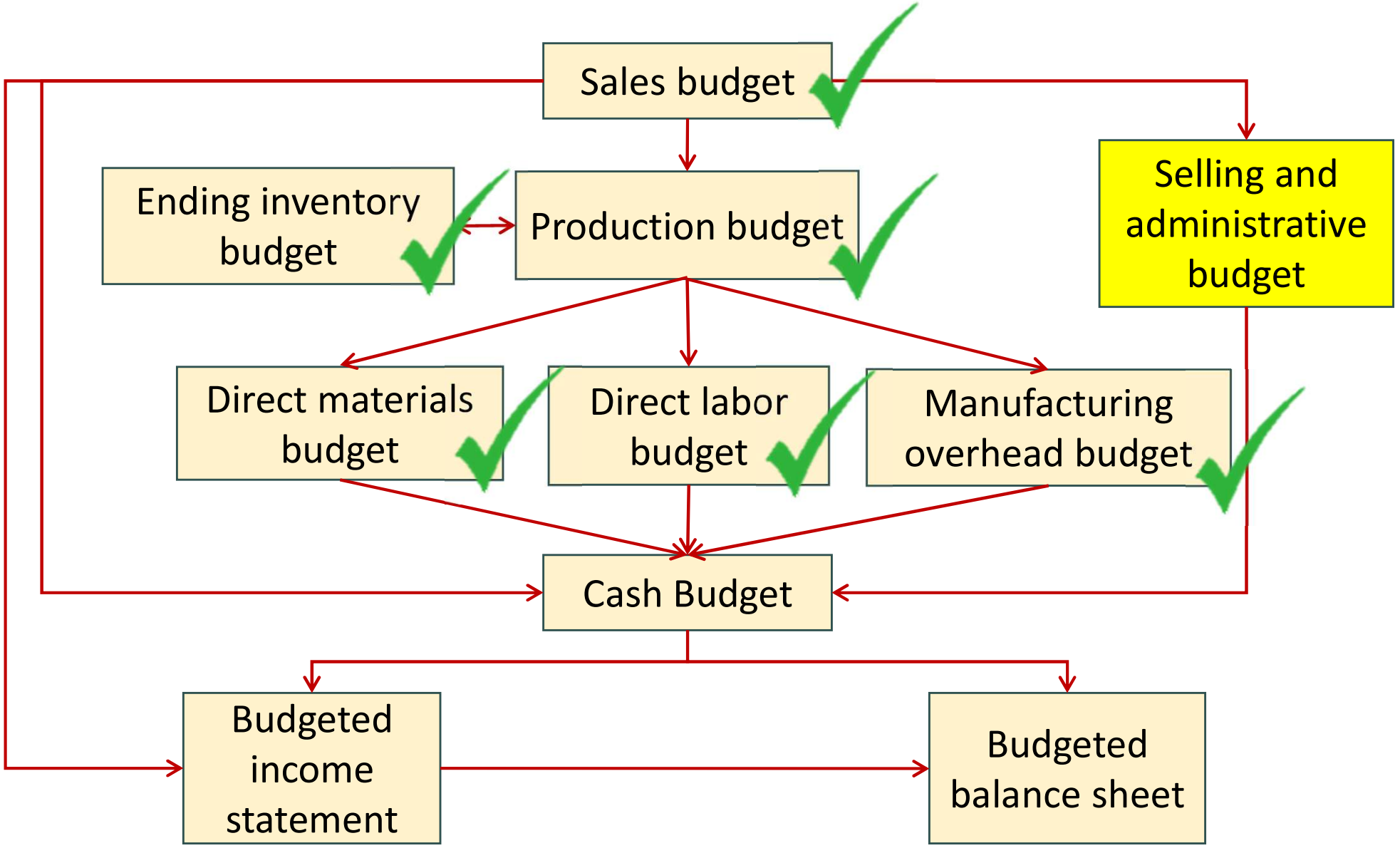
- *How many units left?*
- *How much does each unit cost?*

Ending Finished Goods Inventory Budget - Example

	Calculations	***Per unit***
Direct materials		
Direct labour		
MOH		
Product cost		

	June / Quarter
End FG units in inventory	
Product cost	
End FG inventory bal.	

The Master Budget



Adapted from © McGraw-Hill Education for ACCT102 (BT)

Learning Objective

Prepare a selling and administrative expense budget.

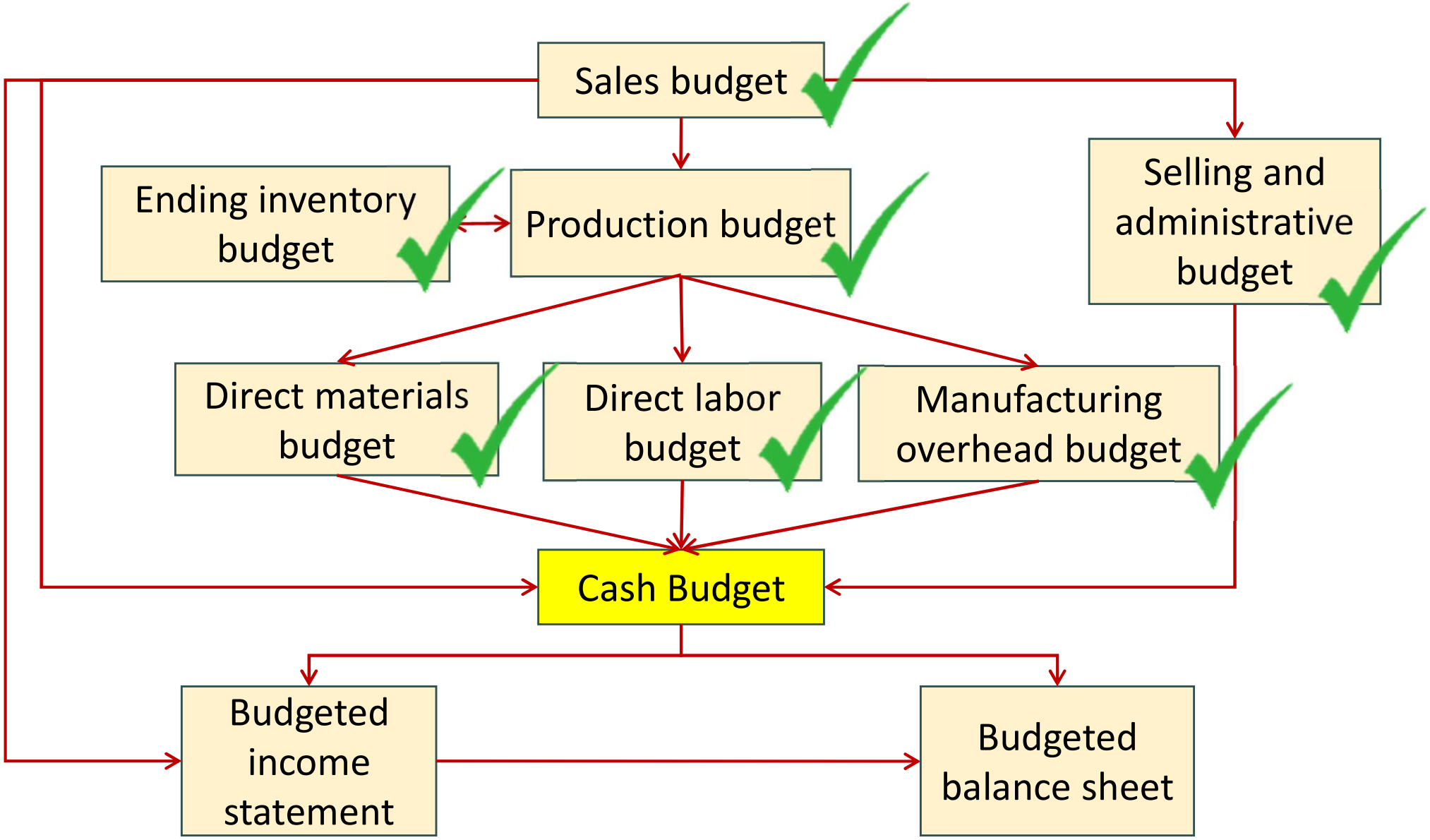
Selling and Administrative Expense Budget - Example

- At Royal, the selling and administrative expense budget is divided into variable and fixed components.
- The variable selling and administrative expenses are $0.50 per unit sold.
- Fixed selling and administrative expenses are $70,000 per month.
- Fixed selling and administrative expenses include $10,000 of depreciation expenses.
- All selling & administrative expenses are paid at the end of the month it is incurred.

Selling and Administrative Expense Budget - Example

	April	May	June	Quarter
Budgeted sales	20,000	50,000	30,000	100,000
Variable S&A rate	0.50	0.50	0.50	0.50
Variable expenses	$ 10,000	$ 25,000	$ 15,000	$ 50,000
Fixed S&A expenses	70,000	70,000	70,000	210,000
Total S&A expenses	80,000	95,000	85,000	260,000
Less: noncash expenses	10,000	10,000	10,000	30,000
Cash S&A expenses	$ 70,000	$ 85,000	$ 75,000	$ 230,000

The Master Budget



Adapted from © McGraw-Hill Education for ACCT102 (BT)

Learning Objective

Prepare a cash budget.

Cash Budget - Example

Beginning cash	***What we have***
Cash receipts	All cash inflows excluding cash received from financing ***What we receive when running our business***
Cash disbursements	All cash payments excluding repayments of principal and interest ***What we disburse when running our business***
Cash excess or deficiency	Determines if the company will need to borrow money or if it will be able to repay funds previously borrowed ***Excess – What we have left*** ***Deficiency – What we do not have***
Financing section	Details the borrowings and repayments projected to take place during the budget period. ***To bridge the liquidity gap and the servicing / repayment of any financing***
Ending cash	***What we ultimately have left / What we wish to have at the end***

Cash Budget - Example

Assume the following information for Royal:

- Maintains a 16% open line of credit of $75,000. Interest is paid quarterly.
- Maintains a minimum cash balance of $30,000.
- Borrows on the first day of the month and repays loans, if there is excess liquidity, on the last day of the quarter.
- Pays a cash dividend of $49,000 in April.
- Purchases $143,700 of equipment in May and $48,300 in June (both purchases paid in cash).
- Has an April 1 cash balance of $40,000.

Cash Budget - Example

	April	May	June	Quarter
Beg. Cash bal.				
Cash collections				
Total cash avail.				
DM				
DL				
MOH				
S&A				
Equipment purchase				
Dividends				
Less: Total disbursements				
Excess/ (Deficiency)				

Cash Budget - Example

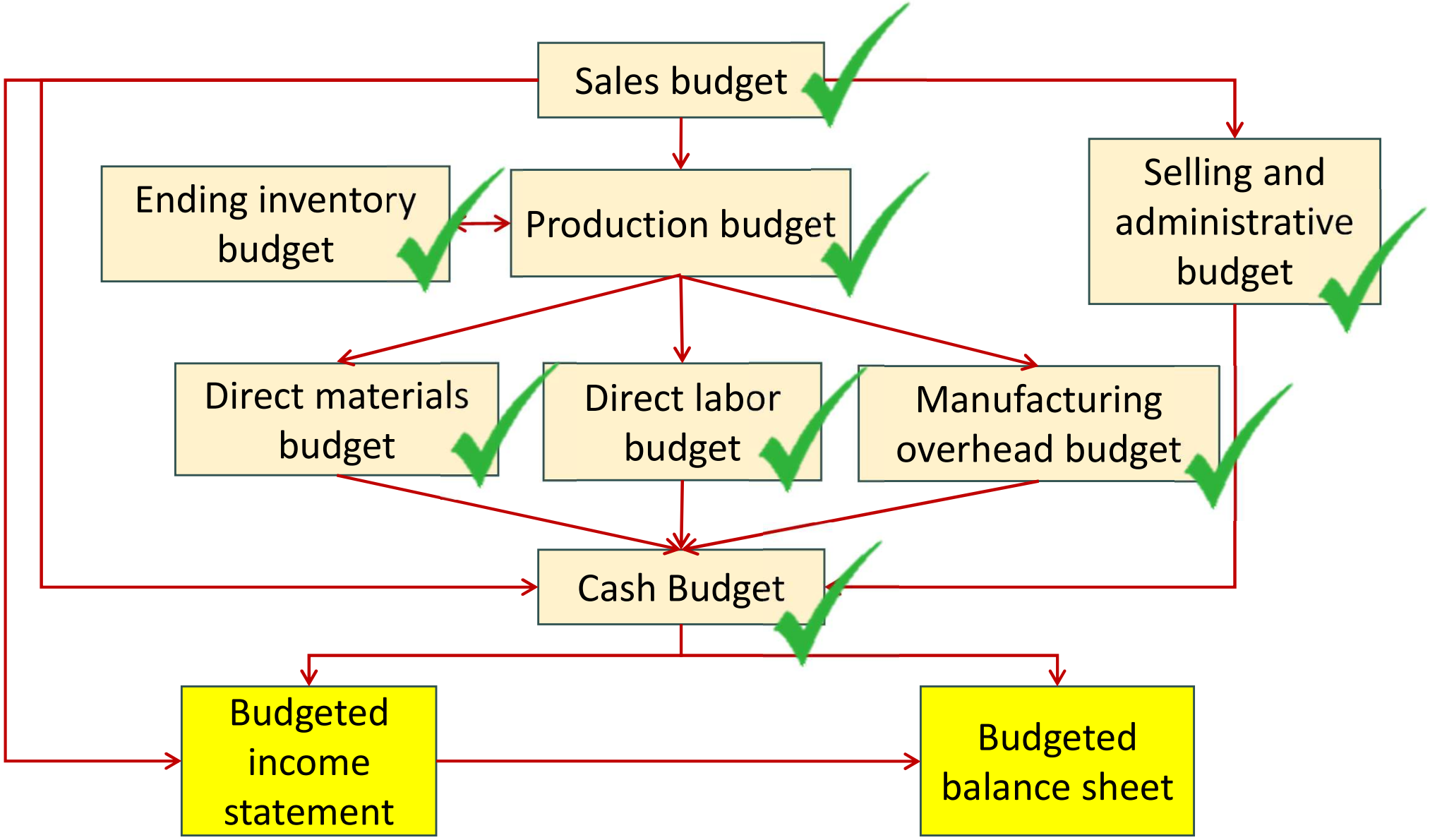
	April	May	June	Quarter
Excess/ (Deficiency)				
Borrowing				
Repayment				
Interest				
Add/(Less): Total financing				
End cash bal.				

Quick Check 10.5

What is the excess (deficiency) of cash available over disbursements for May?

a. $103,500
b. $183,700
c. $60,000
d. $40,000

The Master Budget



Adapted from © McGraw-Hill Education for ACCT102 (BT)

Learning Objective

Prepare a budgeted income statement.

The Budgeted Income Statement

Completed

Cash Budget



Budgeted Income Statement

With interest expense from the cash budget, Royal can prepare the budgeted income statement.

The Budgeted Income Statement - Example

Royal Company Budgeted Income Statement For the Three Months Ended June 30	
Sales	$ 1,000,000
Cost of goods sold	498,515
Gross margin	501,485
Selling and administrative expenses	260,000
Operating income	241,485
Interest expense	1,920
Net income	$ 239,565

Sales budget

Product cost x 100k units sold

S&A budget

Cash budget

The Budgeted Income Statement - Example

Royal Company Budgeted Income Statement For the Three Months Ended June 30	
Sales	$ 1,000,000
Cost of goods sold	498,515
Gross margin	501,485
Selling and administrative expenses	260,000
Operating income	241,485
Interest expense	1,920
Net income	$ 239,565

Learning Objective

Prepare a budgeted balance sheet.

The Budgeted Balance Sheet - Example

Royal reported the following account balances prior to preparing its budgeted financial statements:

- Freehold Land - $50,000
- Common stock - $150,000
- Retained earnings - $158,141 (April 1)
- Property, Plant & Equipment - $175,000

The Budgeted Balance Sheet - Example

Royal Company - Budgeted Balance Sheet As at June 30		
Assets:		
Cash	$	80,580
Accounts receivable		90,000
Raw materials inventory		4,600
Finished goods inventory		24,926
Land		50,000
PPE		277,000
Total assets	$	527,106

Cash budget

30% x $300,000 (June sales)

DM budget $0.40 x 11,500

Ending FG Inventory budget

PPE (April 1)	**$175,000**
Purchase (cash budget)	**$192,000**
Acc depreciation	
MOH budget	**($60,000)**
S&A budget	**($30,000)**
PPE (ending)	**$277,000**

Adapted from © McGraw-Hill Education for ACCT102 (BT)

The Budgeted Balance Sheet - Example

Royal Company - Budgeted Balance Sheet As at June 30	
Labilities & Equity	
Accounts payable	28,400
Common stock	150,000
Retained earnings	348,706
Total liabilities & equity	$ 527,106

50% x $56,800
(June purchase)

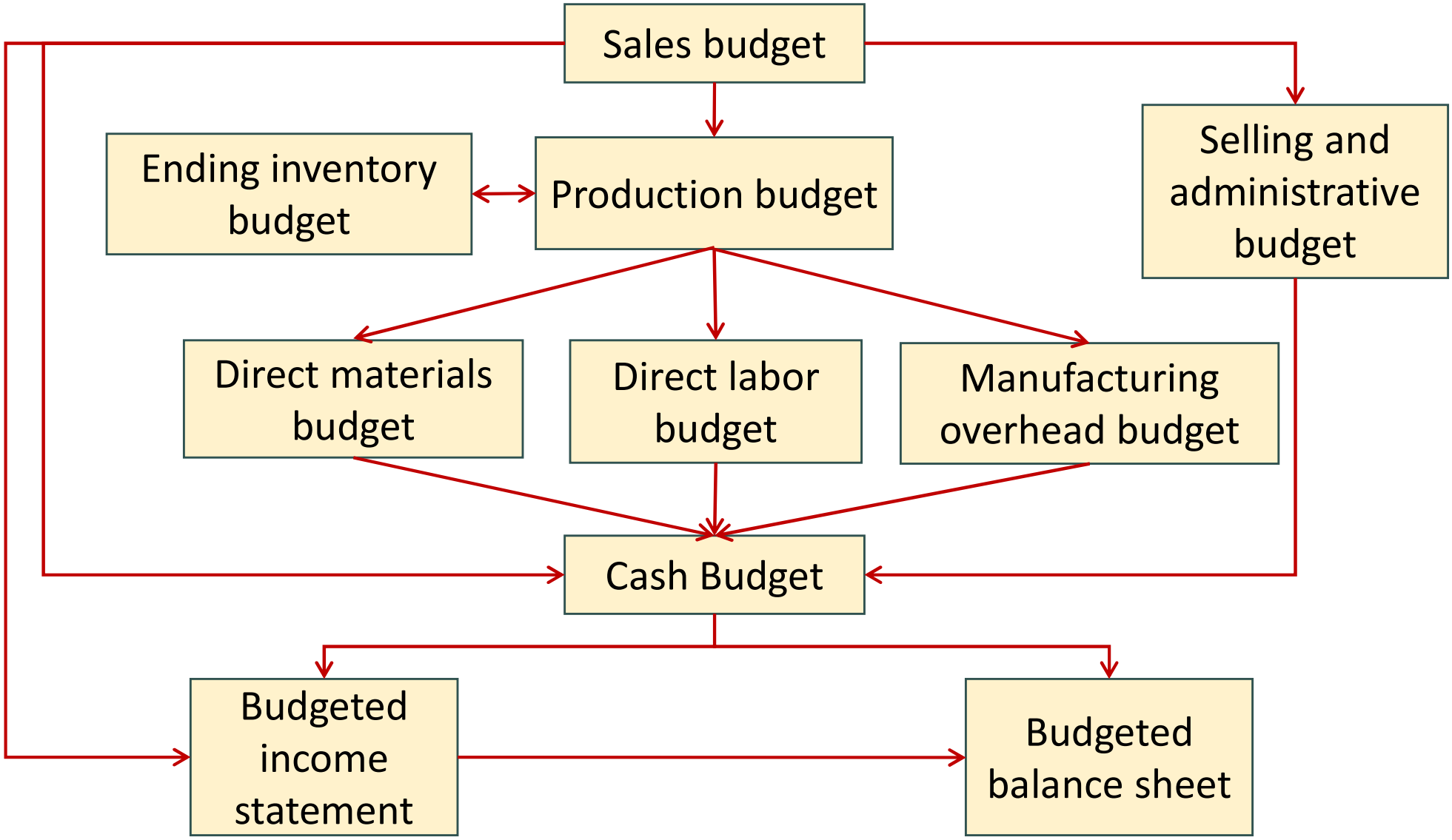
RE (April 1)	**$158,141**
Net income	**$239,565**
Dividends	**($49,000)**
RE (ending)	**$348,706**

The Budgeted Balance Sheet - Example

Royal Company - Budgeted Balance Sheet As at June 30		
Assets:		
Cash	$	80,580
Accounts receivable		90,000
Raw materials inventory		4,600
Finished goods inventory		24,926
Land		50,000
PPE		277,000
Total assets	$	527,106
Labilities & Equity		
Accounts payable		28,400
Common stock		150,000
Retained earnings		348,706
Total liabilities & equity	$	527,106

Adapted from © McGraw-Hill Education for ACCT102 (BT)

How about other types of companies? Merchandising? Service?



Adapted from © McGraw-Hill Education for ACCT102 (BT)

End of topic